Notes on projects

Class project

- **April 14 : Project proposals due (feedback/approval from instructor shortly after)**
- May 5/7: Project presentations
- May 12: Final project reports due (last day of class)

- **Project scope**: Should be comprehensive, following this outline:

  Problem statement in mechanics/materials or other engineering discipline; dataset collection/generation/use; ML algorithm and code development, along with training, testing and validation; critical discussion of results and putting it into perspective of SOTA/literature; conclusion and impact.

  Report can be the slides presented (perhaps with additional backup/material); should include well-documented code and datasets, and any output/media.

Project template

1. **Problem statement** in mechanics/materials: why the problem is important and whether it is a new study?

2. **Dataset collection/generation/use**: Describe the dataset, e.g. use existing dataset or generate your own dataset (collect from literature or generate with simulations/high-throughput experiments)

3. **ML algorithm and code development**: Submit the code you write (model architecture) and specify the training/testing/validation details (loss curve, metrics)

4. **Results and discussions**: Performance of model; Comparison with SOTA; Generalization; …

5. **Conclusion and impact**: Significance; future directions/research; limitations; technological developments, …

6. **References/citations**

Datasets that may be useful

1. Mechanics:

Continuum-level:

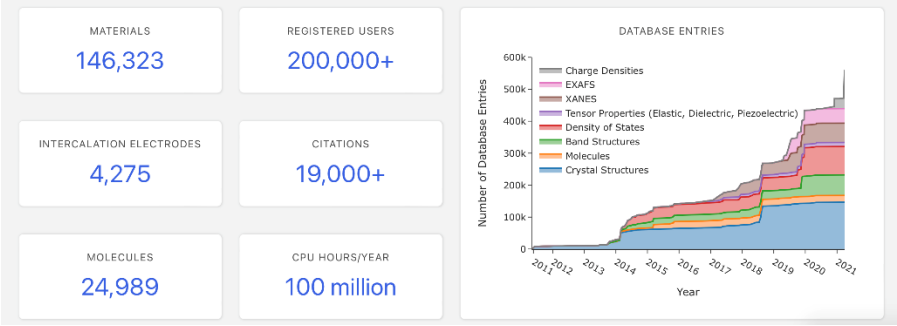
1. **Mechanical MNIST (subsets below):** https://open.bu.edu/handle/2144/39371
    1. Uniaxial Extension
    2. Multi-fidelity
    3. ...
2. **Metamaterials dataset:**
    1. 3D isosurface unit cell & 2D/3D meta-class unit cell: https://decode.mit.edu/datasets/
    2. Mechanical Metamaterial Actuators: https://zenodo.org/records/3891249
    3. Viscoelastic Metamaterials: https://zenodo.org/records/3891249
    4. Architectural cells: https://zenodo.org/records/8210037
3. **Microstructure dataset:**
    1. Two-phase microstructure dataset: https://github.com/lamm-mit/FieldPredictorGAN
    2. Open-phase microstructure dataset: https://microstructures.net/datasets.html
    3. Carbon steel micrograph: https://materialsdata.nist.gov/handle/11256/940?show=full

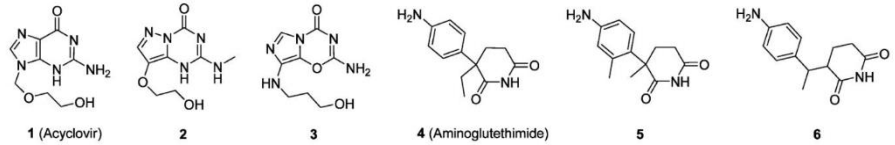
Atomic/molecular-level:

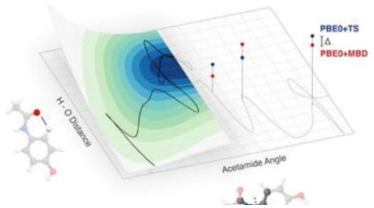
1. **Crystalline solids:**
    1. Strain/stress in Graphene/aluminum nanocrystals: https://github.com/lamm-mit/atomic2field
    2. Crack propagation: https://github.com/lamm-mit/DyFraNet

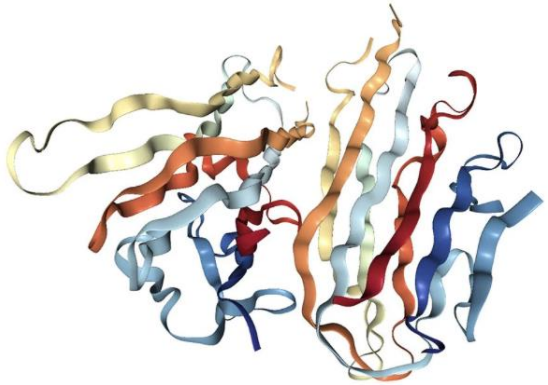
Datasets that may be useful (cont'd)

2. Materials:
   1. Materials Project: https://materialsproject.org
   2. QM7 or QM9: http://quantum-machine.org/datasets/
   3. MD17 (MD trajectory): http://www.sgdml.org/#datasets
   4. MatWeb: https://www.matweb.com/
   5. Protein datasets:
      1. Protein data bank (3D structure): https://www.rcsb.org
      2. UniProt (sequence): https://www.uniprot.org
      3. Protein mechanics: https://github.com/lamm-mit/ProteinMechanicsDiffusionDesign
   6. Nanomaterials datasets:
      1. 2D materials: https://cmr.fysik.dtu.dk/c2db/c2db.html; https://www.materialscloud.org/discover/mc2d/dashboard/ptable
      2. 1D/2D/3D materials:https://statnano.com/nanomaterials









3. Chemistry, e.g. DeepChem

- https://github.com/deepchem/deepchem/releases
- https://deepchem.readthedocs.io/en/latest/api_reference/moleculenet.html

MoleculeNet: A Benchmark for Molecular Machine Learning

Zhenqin Wu,[†,‖] Bharath Ramsundar,[‡,‖] Evan N. Feinberg,[¶,⊥] Joseph Gomes,[†,⊥] Caleb Geniesse,[¶] Aneesh S. Pappu,[‡] Karl Leswing,[§] and Vijay Pande[*,†]

†*Department of Chemistry, Stanford University*

‡*Department of Computer Science, Stanford University*

¶*Program in Biophysics, Stanford School of Medicine*

§*Schrodinger Inc.*

‖*Joint First Authorship*

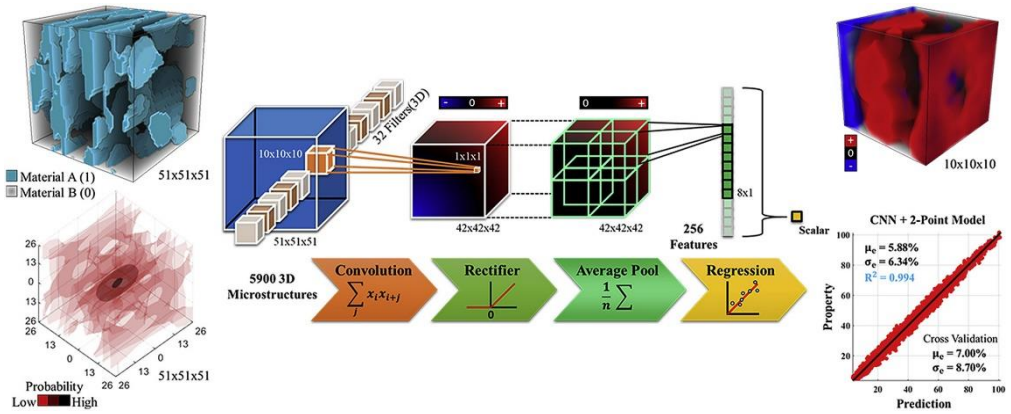
⊥*Joint Second Authorship*

E-mail: pande@stanford.edu

Category	Dataset	Data Type	# Tasks	Task Type	# Compounds	Rec - Split	Rec - Metric
Quantum Mechanics	QM7	SMILES, 3D coordinates	1	Regression	7160	Stratified	MAE
	QM7b	3D coordinates	14	Regression	7210	Random	MAE
	QM8	SMILES, 3D coordinates	12	Regression	21786	Random	MAE
	QM9	SMILES, 3D coordinates	12	Regression	133885	Random	MAE
Physical Chemistry	ESOL	SMILES	1	Regression	1128	Random	RMSE
	FreeSolv	SMILES	1	Regression	642	Random	RMSE
	Lipophilicity	SMILES	1	Regression	4200	Random	RMSE
Biophysics	PCBA	SMILES	128	Classification	437929	Random	PRC-AUC
	MUV	SMILES	17	Classification	93087	Random	PRC-AUC
	HIV	SMILES	1	Classification	41127	Scaffold	ROC-AUC
	PDBbind	SMILES, 3D coordinates	1	Regression	11908	Time	RMSE
	BACE	SMILES	1	Classification	1513	Scaffold	ROC-AUC
Physiology	BBBP	SMILES	1	Classification	2039	Scaffold	ROC-AUC
	Tox21	SMILES	12	Classification	7831	Random	ROC-AUC
	ToxCast	SMILES	617	Classification	8575	Random	ROC-AUC
	SIDER	SMILES	27	Classification	1427	Random	ROC-AUC
	ClinTox	SMILES	2	Classification	1478	Random	ROC-AUC

Possible project ideas (1/4)

- **Topic**: Build structure-property linkage for metamaterials/microstructures:

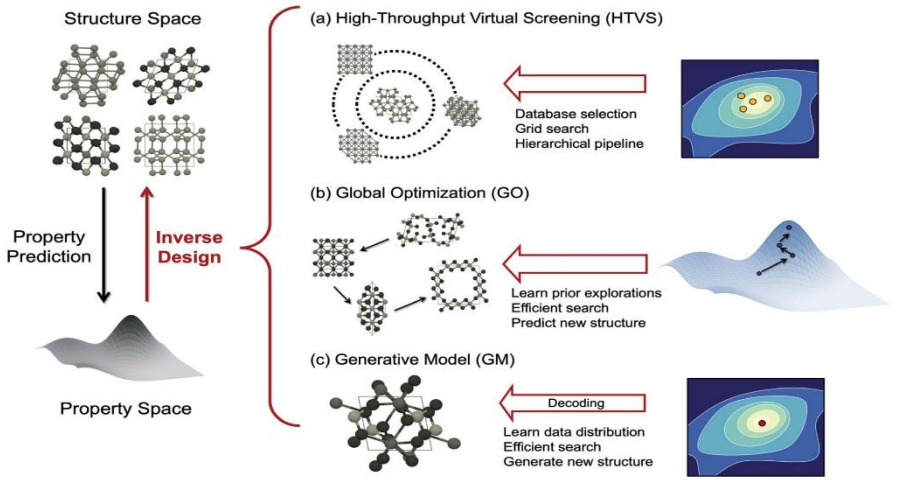


- **Datasets:**
  - Existing ones (e.g. materials project; metamaterials, microstructure dataset) → Compare with SOTA results.
  - Your own dataset (from simulation or experiments: at least hundreds of them) → New results
- **ML models:**
  - Classical ML algorithm (e.g., Bayesian optimization) → small amount of data and simple representation (e.g. vectors)
  - DL models → large amount of data and complicated representation (e.g. CNNs → microstructure images; GNNs → molecules or crystals; language model → proteins)
- **Significance**
  - Acceleration; Transferability; Interpretability; …

Cecen et al. 2018

Possible project ideas (2/4)

- **Topic**: Inverse design of materials/structures using generative methods or combining ML models with optimization algorithms.

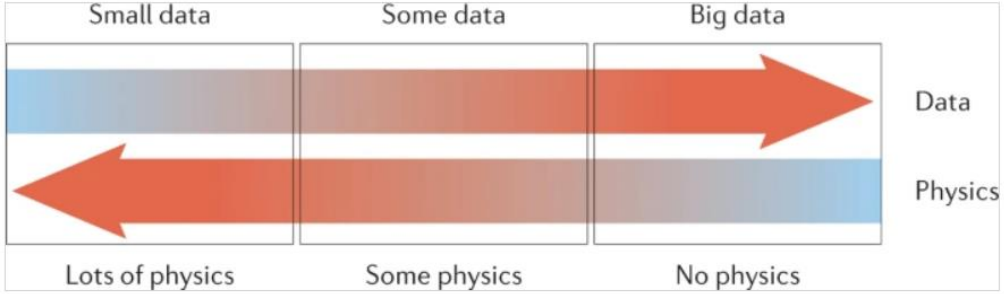


- **Examples and datasets to use:**
  - Design microstructure bypassing strength-ductility tradeoff (Mechanical MNIST, steel microstructure datasets)
  - Design metamaterials (truss-based, cellular, …) with specific functions (e.g. mechanical cloaking, electromagnetic shielding, …)
  - Design protein sequences leading high-strength proteins (BSDB dataset for protein stretching)
  - Design small molecular systems such as chemical compounds (drugs), nanomaterials (graphene, MXene), …
- **ML models:**
  - Generative networks or generative AIs (VAE, GANs, diffusion model, …): add conditions for generating materials with target properties.
  - Combined with optimization algorithm (Genetic algorithm, Bayesian optimization, …): **like we did in Q3 in PSet 2**.
- **Significance**
  - Acceleration; New materials/designs; New mechanism …

Noh et al. 2020

Possible project ideas (3/4)

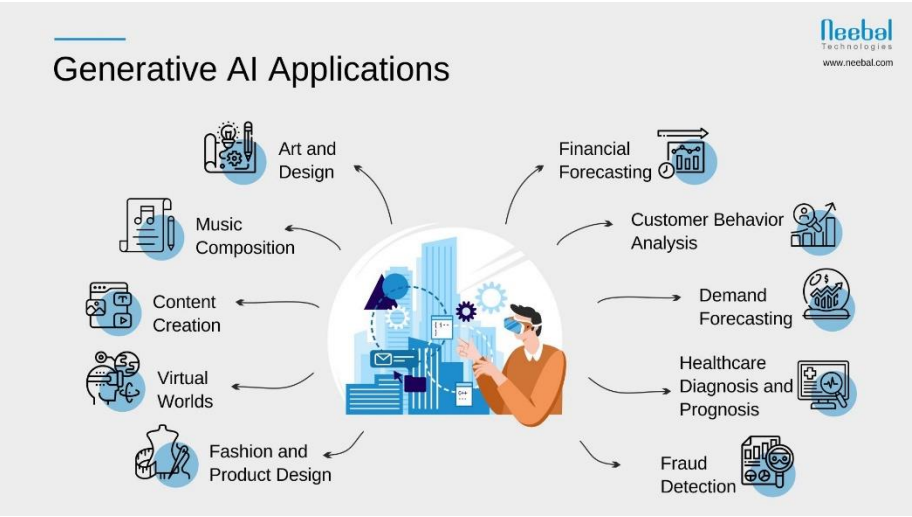
- **Topic**: Encoding prior knowledge of physics for model development or Training models to find underlying physics



- **Examples and datasets to use:**
  - Use PINN to solve catastrophic fracture dynamics (Mechanical MNIST, Fracture dataset: https://github.com/lamm-mit/DyFraNet)
  - Use PINN to calculate strain/stress field (Mechanical MNIST, Composite dataset (https://github.com/lamm-mit/FieldPredictorGAN))
  - Symbolic regression: example: https://arxiv.org/pdf/2303.03192.pdf; Github: https://github.com/WassimTenachi/PhySO.
- **ML models:**
  - Physics-informed neural network (https://maziarraissi.github.io/PINNs/);
  - Hamiltonian neural network (https://greydanus.github.io/2019/05/15/hamiltonian-nns/);
  - Lagrangian neural network (https://greydanus.github.io/2020/03/10/lagrangian-nns/);
  - NNs with invariance or equivalence: e3nn (https://e3nn.org); SE(3)-Transformer (https://fabianfuchsml.github.io/se3transformer/); …
- **Significance**
  - Small data; Discovery of new physics; …

Possible project ideas (4/4)

- **Topic**: Using LLMs or Generative AIs for materials and mechanics problem



- **Examples and resources to use:**
  - Use LLM to mine text, data, …
  - Use Stable Diffusion or DALL-E 2/3 to generate images, or pairs of image-text data
  - Use as tool for question answering, hypothesis generation
  - Multi-modal learning to incorporate different formats of academic resources (text, image, ...) or different aspects of materials (composition, structure, …)
- **Resources:**
  - Open-source model like Llama 2, or others
  - OpenAI AP
  - Hugging Face tutorials: ...
- **Significance**
  - Collect data in an efficient way; build specialized assistant, generate new designs, e.g. microstructures